Haze Thickness Prior-Injected Spectral-Spatial Dehazing Network for Hyperspectral Image

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***Abstract*—Haze pollution leads to reduced spatial visibility and spectral distortion in remote sensing hyperspectral images (HSIs), reducing application capabilities. The distribution of haze in HSIs is spatially irregular and spectrally inhomogeneous, and the effective separation of the haze and the scene information is challenging. Current prior-based methods exhibit weak generalization, and many learning-based approaches introduce irrelevant redundant information, resulting in high computational costs and inadequate utilization of negative samples. To this end, we explore the potential of prior knowledge-guided learning of dehazing networks and propose a haze thickness prior (HTP)-injected spectral-spatial dehazing network for HSI. In particular, by analyzing the difference in haze sensitivity in the visible and infrared, the HTP and prior fusion attention module (PFAM) are proposed to bootstrap the learning of multiple layers of the network, thus enhancing the haze perception capability of the network. We aim to design the spectral-spatial dehazing module (SSDM), which enables the extraction of intrinsic features related to dehaze and reduces the redundancy of features and parameters. In addition, we introduce a multilevel contrastive regularization (MCR) combined with perceptual loss. By explicitly exploiting feature differences across multiple levels between positive and negative samples, this constraint guides the network to learn more discriminative representations and improves dehazing performance. Experimental results on the Gaofen-5 HSI dataset demonstrate that HTP-SSDN has efficient performance that outperforms state-of-the-art methods.**

***Index Terms*—Contrast regularization, dehazing, haze thickness prior (HTP), hyperspectral image (HSI), network.**

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I. Introduction

HAZE represents a prevalent atmospheric phenomenon that significantly degrades machine vision and optical remote sensing imaging capabilities. Hyperspectral imaging sensors, operating across the visible to infrared spectral range, acquire hyperspectral images (HSIs) containing hundreds of narrow, continuous spectral bands, enabling precise target identification and change detection [1], [2]. However, haze-induced effects typically cause reduced spatial visibility and substantial spectral feature distortion in HSIs, ultimately compromising the accuracy of surface feature identification and classification [3], [4]. Consequently, the decoupling of haze-scene information in HSIs, coupled with accurate scene reconstruction and image quality enhancement, becomes imperative to preserve the intrinsic advantages of hyperspectral imaging.

Conventional dehazing methods typically rely on prior knowledge to estimate haze transmittance and atmospheric light, utilizing atmospheric scattering models to restore clean images [5], [6]. These approaches have been successfully applied to both natural and remote sensing images [7], [8]. Due to differences in imaging mechanisms, the dark-object subtraction model has been widely adopted for remote sensing vertical imaging, with various techniques proposed to estimate haze components for image recovery [9], [10], [11]. However, these methods often depend on strong assumptions and

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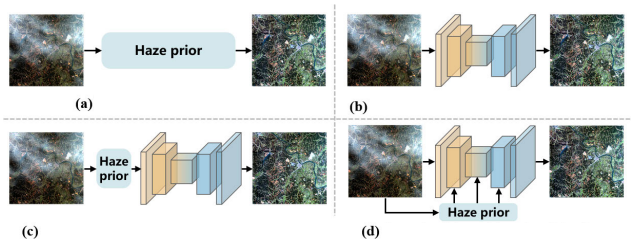


Fig. 1. Comparison of different dehazing methods. Our prior-injected learning dehazing uses the prior information to guide network learning in multiple layers to enhance haze perception and underhaze-scene reconstruction capabilities. (a) Prior dehazing (e.g., DCP). (b) Learning dehazing (e.g., RSDehazeNet). (c) Current prior-learning dehazing (e.g., AU-Net). (d) Our prior-injected learning dehazing.

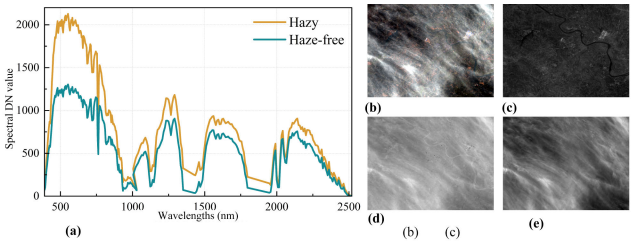


Fig. 2. Observation of the spectral and spatial distribution properties of haze in HSI. Comparison of visible and infrared images, their difference information, and real haze distribution. (a) Spectral profile between hazy and haze-free. (b) Visible wavelength. (c) Infrared wavelength. (d) Difference between (b) and (c). (e) Real haze.

manually tuned parameters, limiting their robustness across diverse scenarios. Alternatively, image processing techniques leveraging intrinsic image features can directly enhance hazy images without requiring model parameter estimation. In the spatial domain, methods such as image enhancement and spatial filtering are employed to accentuate background details while suppressing haze effects [12], [13]. Similarly, in the transform domain, haze and scene components can be separated after appropriate image transformations [14], [15], [16]. As illustrated in Fig. 1(a), these prior-based approaches remain constrained by limited generalizability and suboptimal performance.

Recent advances in deep learning have brought new opportunities for image dehazing, with numerous deep networks proposed to enhance performance [17]. Mainstream approaches employ convolutional neural networks (CNNs) [18], [19], [20], generative adversarial networks [21], [22], and transformers [23], [24], [25] to learn end-to-end mappings from hazy to clean images, thereby bypassing explicit parameter estimation and improving dehazing efficiency [see Fig. 1(b)]. However, these methods heavily depend on the network's ability to perceive haze, and their performance varies due to differences in modeling local versus global features. In addition, the iterative learning required to identify hazy regions increases computational costs. To address these limitations, two-stage dehazing methods [26] have emerged, combining prior knowledge with deep networks [27], [28]. For instance, some approaches first estimate an initial clean image using the atmospheric scattering model (e.g., AU-Net [29]), then refine it via a deep network [see Fig. 1(c)]. However, these methods treat prior information and network learning as separate steps, failing to integrate meaningful haze priors into the neural framework. Recognizing this, we attempt to inject haze prior information directly into the network to guide learning, as illustrated in Fig. 1(d).

Compared to natural and multispectral images (MSIs), haze distribution and removal in HSIs present greater challenges for three key reasons. First, due to the high-spectral dimensionality, haze contaminates most bands and exhibits spectral nonuniformity, with short-wave bands being more haze-sensitive than long-wave bands [see Fig. 2(a)]. Second, haze demonstrates spatial irregularity, as illustrated in Fig. 2(b). Finally, both HSI data and network features suffer from spatial-spectral redundancy, further compounded by dense model parameters [30]. While current HSI dehazing research remains limited, some studies attempt to address these issues through spectral band grouping [31], band selection [32], and attention mechanisms [33]. However, these approaches still exhibit insufficient haze perception capability and struggle to extract low-redundancy spatial-spectral-refined features.

In addition, existing HSI dehazing networks rely solely on clean images as ground truth, employing conventional L1/L2 losses for network optimization. However, this paradigm often results in reconstructed images with compromised detail fidelity and spectral distortion [34], [35]. Although regularization techniques and contrastive learning, which leverage relationships between positive and negative samples, have demonstrated success across various vision tasks, current HSI dehazing methods fail to incorporate hazy images as constraints in the optimization process. Recognizing this, we attempt to improve the performance of HSI dehazing by jointly leveraging clean HSI data as an upper bound and hazy HSI data as a lower bound.

To fully leverage haze prior information for guiding network learning while eliminating redundant information and incorporating regularization constraints, this article proposes a haze thickness prior-injected spectral-spatial dehazing network (HTP-SSDN) for remote sensing HSI dehazing. In HTP-SSDN, the differential image [see Fig. 2(c)], derived from visible and infrared bands, serves as the initial haze before steer network learning via the haze thickness prior (HTP) and the prior fusion attention module (PFAM). The spectral and spatial features are refined and deredundant through a spectral-spatial dehazing module (SSDM), while contrastive regularization is introduced to enhance the network's learning capability. The main contributions of this work are summarized as follows:

1) We propose a novel prior knowledge-injected deep dehazing network (HTP-SSDN) that redefines the utilization of haze prior in HSI dehazing. Rather than passing the prior as a static input, our collaborative HTP and PFAM modules execute a dynamic, multilayer injection of haze statistics across latent representations. The HTP component forces the primary dehazing branch to utilize residual learning to precisely model irregular haze distributions while enabling synchronous prior

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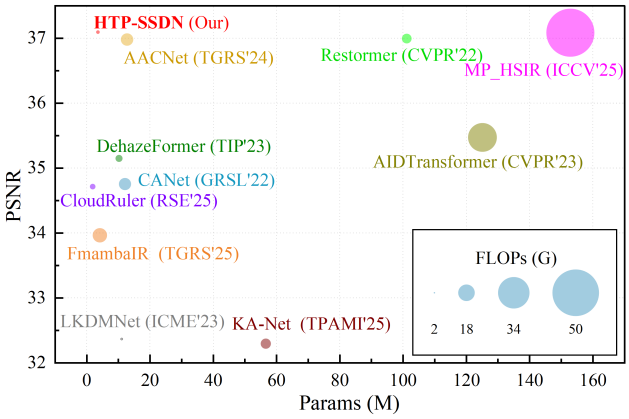


Fig. 3. Comparison of dehazing complexity and performance between HTP-SSDN and other state-of-the-art networks on Gaofen-5 HSI (with size 64 × 64 × 305). The size of the dots indicates the FLOPs of the methods.

updating, thereby enhancing the haze-awareness and reconstruction capabilities of the network.

2) We introduce a feature constraint mechanism tailored to the intrinsic redundancy and intersample characteristics of HSI data. Internally, the introduced SSDM explicitly eliminates intrasample spectral-spatial redundancy to distill core and representative structural features. Externally, this refined representation is cross-constrained by a multilevel contrastive learning scheme that models intersample relationships via positive and negative pairs. By coupling internal structural deredundancy with external cross-sample boundary regularization, the network effectively suppresses spectral artifacts and achieves superior scene generalization.
3) Extensive experiments on both synthetic benchmarks and challenging real-world Gaofen-5 (GF-5) hyperspectral datasets demonstrate that the proposed HTP-SSDN method outperforms state-of-the-art natural and remote sensing image dehazing approaches while maintaining computational efficiency (see Fig. 3), demonstrating prominent practical deployment utility for diverse remote sensing downstream applications.

II. Related Works

A. Prior-Based Dehazing Methods

Prior-based dehazing methods typically rely on physical models such as the atmospheric scattering model and dark-object subtraction model, where mathematical statistics or image transformations are employed to derive useful prior information for efficient parameter estimation. Commonly adopted priors include the dark channel prior (DCP) [5], color attenuation prior [6], local minimal–maximal value prior [36], haze thickness estimation [7], and haze optimized transformation [9]. In HSI dehazing, Kang et al. [37] proposed fog intensity and abundance estimation for haze composition analysis (defog), while Tang and Lin [38] further optimized the haze composition using ADMM-Adam theory. Nevertheless, these methods remain constrained by their inherent assumptions and scenario-specific designs, resulting in limited generalization capability for complex haze conditions.

In practical applications, the extracted haze distribution prior [see Fig. 2(d)] often deviates from actual haze conditions [see Fig. 2(e)]. The haze prior, as shown in Fig. 2(d), retains residual scene information, and its direct application for haze removal typically introduces scene distortion [4]. The effective utilization of such prior information remains challenging. Our approach employs the haze prior as a learning guide, enabling the network to identify haze regions while dynamically updating the prior information precisely.

B. Learning-Based Dehazing Methods

Learning-based dehazing methods generally employ data-driven deep networks to establish nonlinear mappings, shifting from traditional model parameter estimation to direct end-to-end transformation of hazy into clean images [39], [40].

Early research primarily leveraged the image encoding capabilities of CNNs to improve dehazing performance through the incorporation of spatial and channel attention mechanisms, as exemplified by the feature fusion attention network (FFANet) [19], the remote sensing dehazing network (RSDehazeNet) [20], and the channel attention network (CANet) [41]. The adoption of large-kernel convolutions (LKDMNet) [42] further enhanced these approaches. However, constrained by the limited receptive field of CNNs, recent studies have shifted toward transformer and Mamba architectures, capitalizing on their superior global and long-range modeling capacities. [43]. Notable implementations include Restormer [44] for general image dehazing, DehazeFormer [23] for optimized haze removal, and CloudRuler [45] and FmambaIR [46] specifically designed for remote sensing applications, all demonstrating state-of-the-art performance. Despite these advances, the quadratic computational complexity inherent to self-attention mechanisms remains a significant bottleneck.

In HSI dehazing, several deep learning approaches have emerged. Ma et al. [31] developed a spectral group dehazing network (SGNet) to effectively process high-dimensional band information. Xu et al. [33] introduced an asymmetric attention convolutional network (AACNet) for joint feature extraction and haze removal, while Liu et al. [47] proposed a haze density awareness dehazing network that integrated a pretrained spectral angle mapping (SAM) to provide semantic guidance. Wu et al. [48] proposed a multiprompt-guided HSI restoration framework (MP_HSIR) and applied it to the dehazing task. More recently, the field has witnessed innovative applications of unsupervised generative networks [49] and the emerging Mamba [50] architecture for dehazing tasks, indicating promising research directions for computational efficiency and unsupervised learning paradigms.

These approaches typically employ hazy and haze-free image pairs for network training, yet they primarily rely on positive sample constraints while discarding valuable information from negative samples. This limitation often results in artifacts or suboptimal performance. Furthermore, current methods predominantly depend on the network's intrinsic feature extraction capability without effectively incorporating available prior image information, consequently introducing greater uncertainty into the learning process. Our approach synergistically combines haze priors with deep networks for

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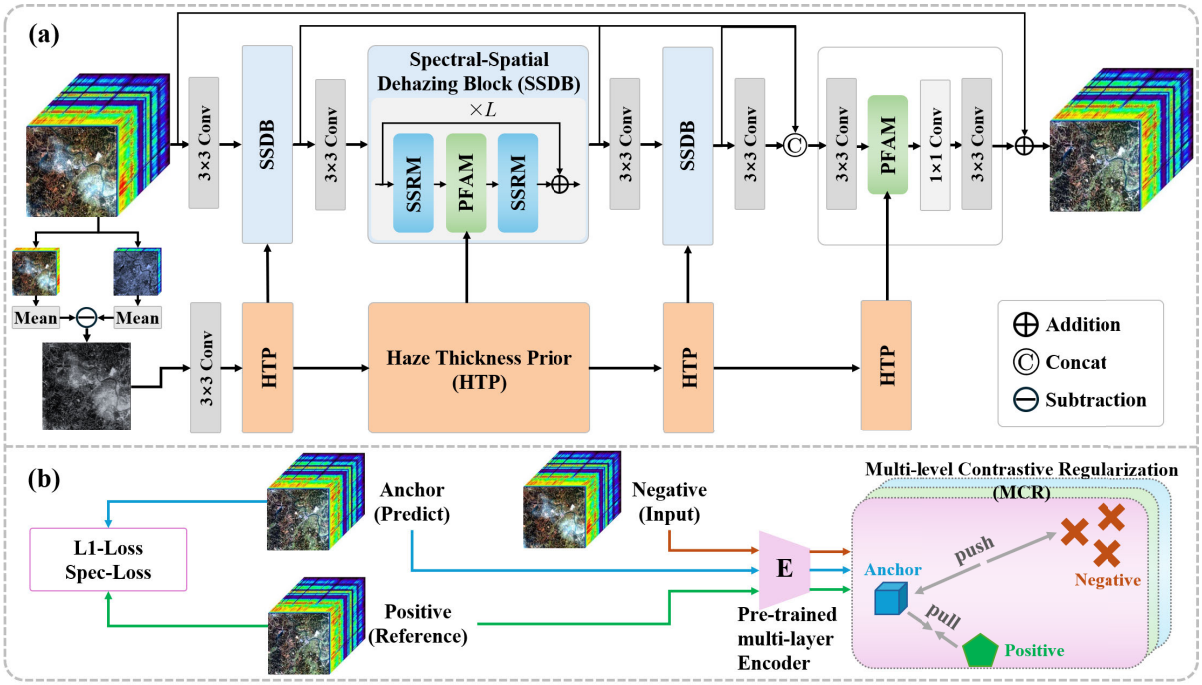


Fig. 4. Illustration of (a) HTP-SSDN framework containing SSDB and HTP with (b) MCR for HSI dehazing.

effective dehazing and scene recovery, while dual-sample constraints (hazy/haze-free) enhance feature representation.

III. Method

A. Overview

In this section, we present the proposed HTP-SSDN framework, as illustrated in Fig. 4. The HTP-SSDN architecture comprises three key components: 1) an end-to-end image dehazing branch; 2) an HTP information extraction and injection branch; and 3) a multiple constraint loss function. The HTP branch primarily performs deep haze feature extraction through convolutional operations applied to visible and infrared differential images (see Section III-B). The dehazing branch, constructed using our novel spatial-spectral dehazing block (SSDB) (see Section III-C), effectively removes haze by integrating HTP information and employing global residual learning. Each SSDB incorporates two essential modules: the spectral-spatial restoration module (SSRM) and the PFAM. Finally, we employ a comprehensive loss function combining L1 norm regularization, spectral constraint, and multilevel contrast regularization (see Section III-D) to optimize the network's performance. This multicomponent loss function ensures robust training and enhanced dehazing results.

B. Haze Thickness Prior

Since haze typically exhibits nonuniform spatial distribution, comprehensively characterizing its distributional properties provides crucial guidance for dehazing. Leveraging the differential sensitivity of haze in visible and infrared images [4], [31], [37], we first extract preliminary difference information to obtain deep haze-distribution features. This extracted prior information is then utilized to guide the dehazing branch for more effective haze removal.

In particular, for a given hazy HSI $\mathcal{X} \in \mathbb{R}^{W\times H\times B}$, where W and H are rows and columns, and B is the number of bands, the average bands in the visible (0.36–0.65 μm) and infrared (2.01–2.36 μm) bands are computed separately, and the initial hazy distribution prior is obtained by differencing the two bands

$$\mathcal{H} = \mathcal{I}_{vis} - \mathcal{I}_{inf} \in \mathbb{R}^{W\times H} \tag{1}$$

where $\mathcal{I}_{vis}$ denotes the average band in the visible, which contains the haze distribution as well as the scene information, and $\mathcal{I}_{inf}$ denotes the average band in the infrared, which contains the scene information that is less disturbed by the haze. This band selection is based on Rayleigh and Mie scattering theories, where shorter visible wavelengths (0.36–0.65 μm) are heavily scattered by aerosols, while longer short-wave infrared (SWIR) wavelengths (2.01–2.36 μm) easily penetrate haze to preserve clear surface details. In scenes dominated by urban structures, bare soil, roads, and water bodies, where the visible-infrared contrast remains flat and stable, HTP demonstrates high reliability. However, it is prone to misidentifying white snow textures as haze over landcovers with sharp spectral transitions, such as snow. Regarding sensor sensitivity, HTP exhibits inherent robustness rather than dependence on specific spectral channels. Because HTP uses averaged information across continuous spectral ranges rather than single-channel inputs, migrating to other hyperspectral sensors (e.g., AVIRIS, PRISMA, or EnMAP) allows equivalent continuous visible and SWIR regions to be seamlessly generalized.

After the calculation of the difference, the preliminary haze prior image $\mathcal{H}$ contains more haze components and can

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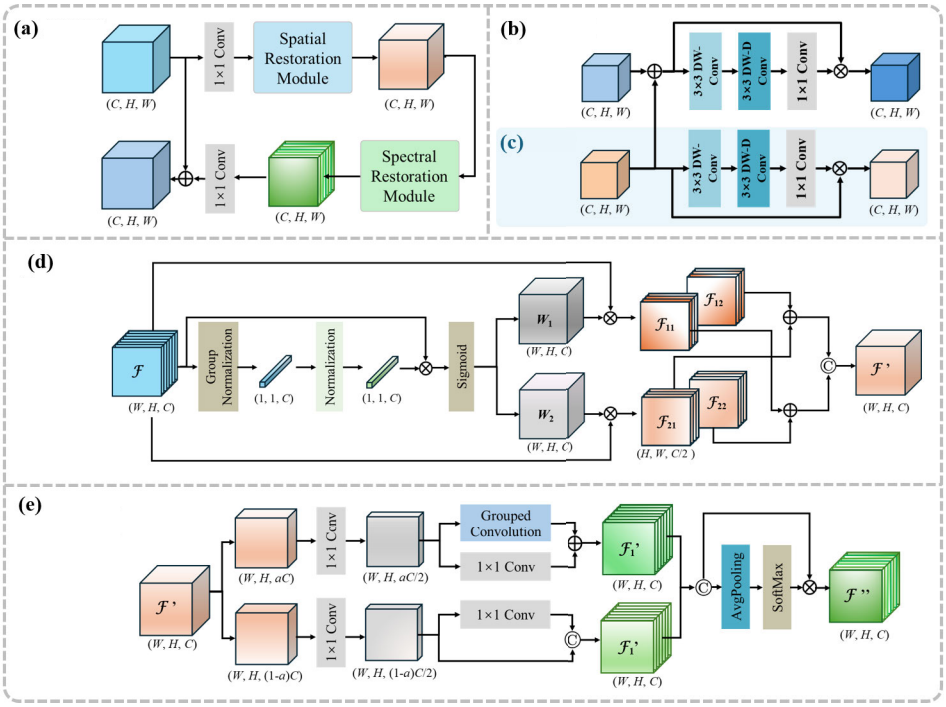


Fig. 5. Illustration of (a) SSRM containing (d) SpaRM and (e) SpeRM, and (b) PFAM, of which (c) HTP is a subbranch.

basically reflect the spatial distribution of haze. However, some of the residual scene information in $\mathcal{H}$ will affect the accurate characterization of haze. Therefore, we adopt a 3×3 convolutional layer to extract the deeper haze-distribution feature $\mathcal{X}_{H0} \in \mathbb{R}^{W \times H \times C}$ [C is the number of feature channels] and use it as the input to the HTP.

The HTP module corresponds to the lower branch of the two parallel branches in PFAM [see Fig. 5(c)]. It takes the shallow feature $\mathcal{X}_{H0}$ as input; one output is fused with PFAM's upper branch (see Section III-C-PFAM), while the other is processed by a 3-D attention composed of a 1×1 convolution, a depthwise dilated convolution, and a depthwise convolution [51]. This produces refined haze-distribution features that are passed to the next HTP module. This process can be expressed as

$$Att = \text{Conv}_{1\times1}\left(\text{DWDConv}\left(\text{DWConv}\left(X_{Hi-1}\right)\right)\right) \tag{2}$$

$$\mathcal{X}_{Hi} = \mathcal{X}_{Hi-1} \otimes Att, i = 1, 2, \ldots \tag{3}$$

where $\text{Conv}_{1\times1}$, DWDConv, and DWConv denote 1×1 convolution, depthwise dilation convolution, and depthwise convolution, respectively. $\mathcal{X}_{Hi-1}$ denotes the input features of the previous layer of HTP, and $\mathcal{X}_{Hi}$ denotes the output refined features from HTP, $Att \in \mathbb{R}^{W \times H \times C}$ denotes the attention map, $\otimes$ means elementwise product.

The multilayer HTP-derived haze prior provides essential guidance to the dehazing branch, enabling precise characterization of haze distribution and ultimately enhancing the dehazing performance.

C. Spectral-Spatial Dehazing Block

Haze exhibits nonuniform distributions across both spectral and spatial domains, necessitating focused attention on critical haze regions and band-specific characteristics to facilitate effective image reconstruction. However, HSIs and their convolutional features typically contain substantial spectral-spatial redundancies. Effectively constraining these redundant features plays a crucial role in image reconstruction by simultaneously reducing model parameters while enhancing feature discriminability [30]. To this end, we propose SSDM to extract the key features used for feature characterization, which consists of a combination of L SSRM-PFAM-SSRM and a skip connection [see Fig. 4(a)].

SSRM: SSRM utilizes a tandem composition of a 1×1 convolution, a spatial restoration module (SpaRM), a spectral restoration module (SpeRM), and 1×1 convolution [see Fig. 5(a)], which is mainly used to reduce data redundancy and extract key information.

For the SpaRM, its main idea is separation and reorganization. Separation is the division of features with different spatial informativeness [see Fig. 5(d)]. In particular, for a given feature map $\mathcal{F} \in \mathbb{R}^{W \times H \times C}$, the information content of different feature maps is first evaluated by using the scaling factor of the group normalization layer [51], which is a process that normalizes the input features by subtracting the mean and dividing it by the standard deviation and yields the trainable parameter vector $\delta \in \mathbb{R}^C$ to measure the spatial variance of pixels within each batch and channel. Then, the normalized correlation weights $W_\delta \in \mathbb{R}^C$ of different feature maps are calculated using δ as follows:

$$W_\delta = \frac{\delta_i}{\sum_{i=1}^{C} \delta_i}, i = 1, 2, \ldots, C. \tag{4}$$

The obtained weights W_δ are weighted on the input feature $\mathcal{F}$ and mapped to the range (0, 1) by a sigmoid function. Finally, we define a threshold $T = 0.5$ and set the weights

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above the threshold to 1 to obtain the informative weights W_1, and the weights below the threshold to 0 to obtain the noninformative weights W_2. The input feature $\mathcal{F}$ is multiplied with weights W_1 and W_2, to obtain the informative feature $\mathcal{F}_1$ and the less informative feature $\mathcal{F}_2$, which enables the separation of different spatial informative features.

Reorganization is a combination of features with different information content to generate rich and more informative features. In particular, we partition the obtained features $\mathcal{F}_1$ channelwise into subsets $\mathcal{F}_{11}$ and $\mathcal{F}_{12}$, and similarly partition $\mathcal{F}_2$ into $\mathcal{F}_{21}$ and $\mathcal{F}_{22}$. Then, elementwise summation is performed between subsets $\mathcal{F}_{11}$ and $\mathcal{F}_{22}$, $\mathcal{F}_{12}$ and $\mathcal{F}_{21}$. Through cross-summation features from the two subsets, the information-rich features are fully fused with the less informative ones, thereby enhancing the information flow between them. Subsequently, the two cross-summation features are concatenated to obtain the spatially refined feature map $\mathcal{F}'$. The process can be represented as follows:

$$\mathcal{F}' = \text{Concat}\left(\left(\mathcal{F}_{11} \oplus \mathcal{F}_{22}\right), \left(\mathcal{F}_{12} \oplus \mathcal{F}_{21}\right)\right) \tag{5}$$

where $\oplus$ represents elementwise summation and Concat$(\cdot)$ denotes the concatenation of two tensors. SSRM can enhance the spatial representation of haze and eliminate redundant features by separating and reorganizing features of different informativeness. Significantly, this strategy is uniquely suited for HSI dehazing because haze disrupts spatial textures nonuniformly across continuous bands. Separating spatial features based on informativeness isolates heavily degraded, low-frequency haze components from high-frequency, clean land-cover details, while cross-reorganization ensures robust information flow and prevents spectral distortion during texture reconstruction.

For the SpeRM, it has a similar idea to the SpaRM, i.e., to further reduce feature redundancy by separating and reorganizing the spatial-refined features in the spectral direction [see Fig. 5(e)]. In particular, given the feature $\mathcal{F}'$, a separation coefficient $0 \le \alpha \le 1$ is first introduced to divide channels C into a weighted combination of αC channels and $(1 - \alpha)C$ channels. The spectral features are further optimized by using a 1×1 convolution to obtain the features $\mathcal{F}'_{\alpha C}$ and $\mathcal{F}'_{(1-\alpha)C}$, respectively. Then, $\mathcal{F}'_{\alpha C}$ is fed into a dual-channel processor with grouped convolution (group size = 2) and a 1×1 convolution, and the two output features are summed to complete the extraction of highly informative spectral representative and reduce the computational parameters, which can be expressed as

$$\mathcal{F}'_1 = \text{GConv}\left(\mathcal{F}'_{\alpha C}\right) \oplus \text{Conv}_{1\times 1}\left(\mathcal{F}'_{\alpha C}\right) \tag{6}$$

where $\mathcal{F}'_1 \in \mathbb{R}^{W\times H\times C}$ is the obtained enriched spectral features. GConv denotes grouped convolution. Meanwhile, $\mathcal{F}'_{(1-\alpha)C}$ is fed into a 1×1 convolution, and the generated features are concatenated with the input features to obtain the complement of $\mathcal{F}'_1$ features as follows:

$$\mathcal{F}'_2 = \text{Concat}\left(\text{Conv}_{1\times 1}\left(\mathcal{F}'_{(1-\alpha)C}\right), \mathcal{F}'_{(1-\alpha)C}\right) \tag{7}$$

where $\mathcal{F}'_2 \in \mathbb{R}^{W\times H\times C}$ is the obtained complementary spectral feature. In the reorganization stage, we used global average pooling and a softmax function to get the weight vectors while connecting the two spectral features to optimize the spectral information further, and the process can be expressed as follows:

$$W_{spe} = \text{AvgPool}\left(\phi\left(\text{Concat}\left(\mathcal{F}'_1, \mathcal{F}'_2\right)\right)\right) \tag{8}$$

$$\mathcal{F}'' = W_{spe} \otimes \text{Concat}\left(\mathcal{F}'_1, \mathcal{F}'_2\right) \tag{9}$$

where $\mathcal{F}''$ is the final spectral-refined feature, AvgPool$(\cdot)$ is global average pooling, and ϕ denotes softmax function. Through the lightweight convolution operation, the SpeRM reduces the redundant information and the number of parameters while extracting representative spectral features.

PFAM: PFAM aims to incorporate HTP into deep networks to guide detailed reconstruction and haze removal in dehazing networks. In particular, it has two parallel branches, each consisting of a 3-D attention [51]. For the dehazing branch, it adds the features of the upper SSRM with the prior features from the HTP to increase the attention of the haze-covered area; the 3-D weights are obtained by multilayer convolution, which can further improve the haze perception of the features [see Fig. 5(b)], and the process can be expressed as

$$Att' = \text{Conv}_{1\times 1}\left(\text{DWDConv}\left(\text{DWConv}\left(\mathcal{F}''_j \oplus \mathcal{X}_{Hi}\right)\right)\right) \tag{10}$$

$$\mathcal{F} = \left(\mathcal{F}''_j \oplus \mathcal{X}_{Hi}\right) \otimes Att' \tag{11}$$

where $\mathcal{F} \in \mathbb{R}^{W\times H\times C}$ denotes the output feature, $\mathcal{X}_{Hi}$ denotes the prior feature from HTP. The prior branch then optimizes the HTP features while outputting the deeper prior features [(2) and (3)].

D. Multiple Constraints

In image dehazing, L1 loss has been shown to outperform L2 loss in terms of peak signal-to-noise ratio (PSNR) and structural similarity index measurement (SSIM). Therefore, we adopt the L1 loss to improve the similarity between dehazing HSI and clean HSI, which can be expressed as

$$\mathcal{L}_1 = \|\mathcal{Y} - \text{HTPSSDN}\left(\mathcal{X}\right)\|_1 \tag{12}$$

where $\mathcal{Y}$ denotes clean HSI. To assist in HSI spectral profile reconstruction, we further employ the spe loss [4] to enhance spectral reconstruction, which can be expressed as

$$\mathcal{L}_{spe} = \|\mathbf{s}_{\mathcal{Y}} - \text{HTPSSDN}\left(\mathbf{s}_{\mathcal{X}}\right)\| \quad \left(\mathbf{s}_{\mathcal{Y}} \in \mathbb{R}^{1\times B}, \mathbf{s}_{\mathcal{X}} \in \mathbb{R}^{1\times B}\right) \tag{13}$$

where $\mathbf{s}_{\mathcal{Y}}$ and $\mathbf{s}_{\mathcal{X}}$ denote the mean spectral pixel vector of the clean HSI and the hazy HSI.

Nevertheless, the aforementioned loss functions merely evaluate the pixel-level discrepancy between the predicted and clean images, disregarding the valuable information contained in the hazy input, which serves as a natural lower bound. To overcome this limitation, we need to simultaneously maximize the similarity between the dehazed output and the clear reference image while minimizing its similarity to the hazy input, thereby achieving superior restoration. In addition, we aim to define interimage similarity using multilevel high-level features, thereby leveraging semantic and stylistic cues that better reflect perceptual differences. Motivated by contrastive learning [34], [52] and perceptual loss [53], we introduce a

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novel representation scheme that exploits the representational capability of pretrained visual models. By extracting features at multiple depths, our method reduces the semantic distance between the network output and the clear image while enlarging the semantic distance to the hazy image. In particular,

1) From the dehazed output (anchor), clean image (positive), and hazy input (negative), we extract three visible bands from the HSI at 471.625, 548.666, and 638.598 nm, which correspond to true-color surface appearance. These bands capture haze distribution and align with the channel structure expected by pretrained visual models.
2) Leveraging the VGG network's multilevel representational capability used in perceptual loss and the stability of its parameters [53], [54], we adopt it as the feature encoder to implement the multilevel contrastive regularization (MCR) scheme, as illustrated in Fig. 4(b)

$$\mathcal{L}_{MCR} = \sum_{i=1}^{m} \beta_i \frac{\|\mathbf{E}_i(\mathcal{Y}) - \mathbf{E}_i(\mathrm{HTPSSDN}(\mathcal{X}))\|_1}{\|\mathbf{E}_i(\mathcal{X}) - \mathbf{E}_i(\mathrm{HTPSSDN}(\mathcal{X}))\|_1 + \epsilon} \quad (14)$$

where $\mathbf{E}$ denotes pretrained multilayer encoder and $\mathbf{E}_i(\cdot)$ represents the feature output of the ith layer, β_i is the weight coefficient of the ith layer. ϵ is a stabilization parameter to ensure continuous numerical stability during end-to-end training. Finally, we formulate the overall loss function as follows:

$$\mathcal{L}_{total} = \mathcal{L}_1 + \lambda_1 \mathcal{L}_{spe} + \lambda_2 \mathcal{L}_{MCR} \quad (15)$$

where λ_1 and λ_2 are set to 0.01 and 0.001, respectively, for balancing loss terms and enhancing dehazing effectiveness. Parameter analysis is detailed in Section IV-F.

IV. Experiments and Discussions

A. Datasets and Experimental Setup

1) Datasets: We evaluated the effectiveness of the proposed method on two public Gaofen-5 (GF-5) HSI datasets: HyperDehazing [4] and the hyperspectral defogging dataset (HDD) [38].

HyperDehazing[1] is a synthetic hazy dataset, which contains 2000 pairs of hazy and haze-free images; each image has a size of $512 \times 512 \times 305$, with a variety of haze morphologies and distributions. 90% of this dataset was used for network training, while the remaining data was used for testing.

HDD is a real-world hazy dataset, which contains 20 hazy images without reference; each image has a size of $512 \times 512 \times 305$ with high hazy concentration. All images were used for network testing.

2) Implementation Details: The hyperparameters of our method were set as follows. The separation coefficient was $\alpha = 1/2$, and the number of SSRM-PFAM-SSRM in SSDB was $L = 3$, as analyzed in Section IV-F. In MCR, the VGG19 encoder was employed to extract features from five layers for contrastive regularization, with weight coefficients set to $[1/32, 1/16, 1/8, 1/4, 1]$ [34]. The constant is $\epsilon = 10^{-7}$. Driven jointly by the pixel-level loss, the network output steadily converges toward the clear target, thereby forcing the anchor-to-positive distance to zero. Meanwhile, because the physical discrepancy between hazy and clear images is fixed and substantial, the denominator (anchor-to-negative distance) naturally expands and stabilizes at a steady lower bound. Consequently, the loss ratio safely diminishes without risking gradient explosion or numerical singularity across all five VGG layers. According to the experience in [4] and [50], the training data size was cropped to 64×64, the batch size was set to 2, and all datasets were trained for 10 000 iterations. The Adam optimization operator was employed to accelerate the training, where the momentum parameters were set to 0.9, 0.999, and 10^{-8}, respectively. The initial learning rate was set to 1×10^{-4}, with the cosine annealing strategy to adjust the learning rate. The whole network was implemented on the PyTorch framework with an NVIDIA GeForce RTX 3060 GPU.

3) Competitors and Evaluation Metric: We compared our method with the prior-based methods: DCP [5] and defog [38], and learning-based methods; including CNN-based methods: FFANet [19], RSDehazeNet [20], PSMBNet [55], CANet [41], LKDMNet [42], PhDNet [28], SGNet [31], and AACNet [33]; transformer-based methods: Restormer [44], DehazeFormer [23], AIDTransformer [24], CloudRuler [45], and MP_HSIR [48]; mamba-based method: FmambaIR [46]; and a contrast learning-based method: KA-Net [56]. Defog, SGNet, AACNet, and MP_HSIR are the HSI dehazing methods currently available; other thin cloud and haze removal models have been included to ensure the evaluation remains state-of-the-art. To quantitatively assess dehazing performance, SSIM, PSNR, universal image quality index (UQI), and SAM metrics were adopted for paired image evaluation [38], while natural image quality evaluator (NIQE) and average gradient (AG) metrics were adopted for real images [4]. To guarantee a strictly fair evaluation, all learning-based competitors were retrained from scratch and tested under identical matched conditions using the same hyperspectral datasets and hardware configurations.

B. Scene Dehazing Results

HyperDehazing: We validated the performance of different dehazing methods on the HyperDehazing dataset and gave a quantitative analysis (see Table I) and visual comparison (see Fig. 6).

Table I demonstrates the performance of the different dehazing methods, where the learning-based methods were significantly better than the prior-based methods, and the proposed HTP-SSDN had the best metrics. In particular, the prior-based methods generally had low quantitative metrics due to limited generalization performance, and defog performed less than DCP due to excessive dehazing. Among learning-based methods, network models such as FFANet, RSDehazeNet, CANet, and LKDMNet had more limited performance, e.g., the SSIM is lower than 0.96, and the latest KA-Net model for natural images also fell short, struggling to represent haze distribution across complex spectra; the state-of-the-art transformer methods, e.g., Restormer, AIDTransformer, generally had higher performance, achieving an SSIM of around 0.97, CloudRuler achieved lower metrics, likely due to the difficulty of applying physical

[1] https://github.com/RsAI-lab/HyperDehazing

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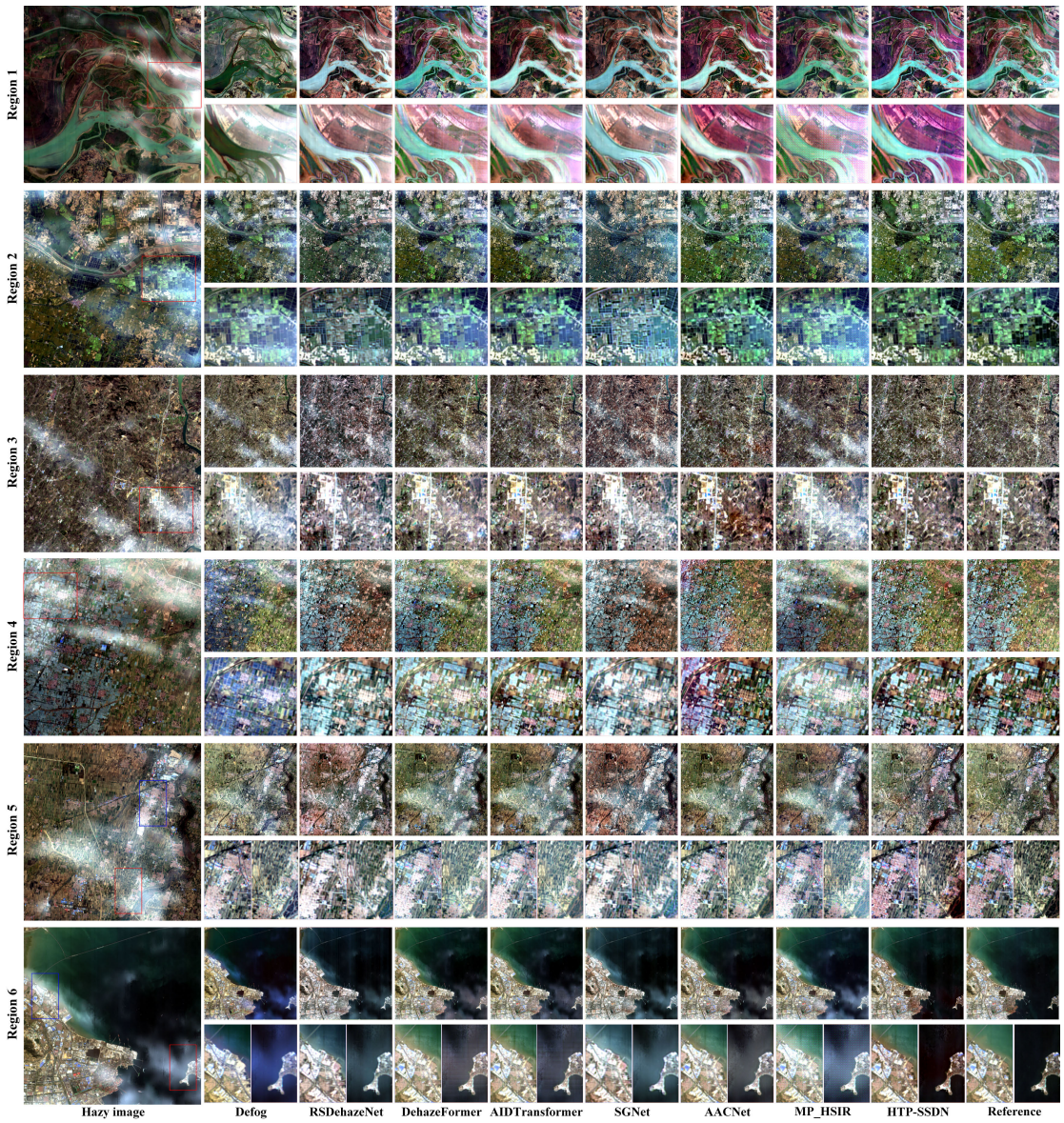


Fig. 6. Dehazing results (Red: band 59, Green: band 38, Blue: band 20) of various methods on the HyperDehazing dataset. For each region, (top row) dehazing result for the whole scene, and (bottom row) zoomed area.

constraints to continuous narrow bands; FmambaIR demonstrated high dehazing accuracy through frequency-domain processing, proving the potential of the frequency domain in HSI restoration; and among the HSI dehazing methods, SGNet achieved the highest UQI, and AACNet and MP_HSIR achieved suboptimal SSIM and PSNR, respectively, demonstrating the effectiveness of spectral and spatial information utilization in dehazing. The proposed method achieved optimal results in all three metrics, which proves its excellent dehazing performance.

Fig. 6 illustrates six regions with inhomogeneous haze distribution, with surface scenes covering wetlands, farmland, mountains, cities, and oceans, with zoomed-in detail comparisons in the red areas. Based on the experimental results, it can be seen that defog suffered from both haze residuals (regions 3 and 6) and spectral aberration problems (regions 4 and 7), and its haze removal and generalization performance were limited. The scenes recovered by RSDehazeNet suffered from more severe spectral aberrations. DehazeFormer and AIDTransformer had incomplete haze removal in most of the scenes (regions 3 and 6). For the HSI dehazing methods, SGNet had poor recovery in agricultural areas (regions 4 and 5), AACNet had limited performance in marine scenes (region 6), and MP_HSIR failed to restore the surface scene, with visually noticeable haze residue remaining. On the contrary, the proposed HTP-SSDN effectively removed haze and recovered the surface scene closest to the haze-free reference.

HDD: Quantitative analysis (see Table II) and visual comparisons (see Fig. 7) of the different dehazing methods were also given on the HDD dataset.

Table II gives the quantitative metrics of dehazing without reference HSI. It can be seen that the learning-based method still outperformed the prior-based methods, and the transformer-based dehazing method outperformed the

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TABLE I

QUANTITATIVE COMPARISON OF DEHAZING METHODS ON THE HYPERDEHAZING DATASET. BOLD AND UNDERLINED INDICATE BEST AND SECOND-BEST RESULTS

Methods		Venue	SSIM↑	PSNR↑	UQI↑	SAM↓
Prior-based	DCP [5]	TPAMI 2009	0.7809	26.2919	0.6906	0.1821
	Defog* [38]	TGRS 2022	0.6119	22.7030	0.4953	0.5777
Learning-based	FFANet [19]	AAAI 2020	0.8955	30.2647	0.9418	0.0900
	RSDehazeNet [20]	TGRS 2021	0.9394	33.9591	0.9759	0.0903
	CANet [41]	GRSL 2022	0.9543	34.7565	0.9754	0.0676
	Restormer [44]	CVPR 2022	0.9700	36.9971	0.9746	0.0428
	SGNet* [31]	ISPRS 2022	0.9661	36.9644	**0.9815**	0.0980
	LKDMNet [42]	ICME 2023	0.9561	32.3696	0.9583	0.0598
	PSMBNet [55]	TGRS 2023	0.9687	35.6614	0.9802	0.0485
	DehazeFormer [23]	TIP 2023	0.9683	35.1518	0.9731	0.0464
	AIDTransformer [24]	CVPR 2023	0.9701	35.4737	0.9737	0.0323
	AACNet* [33]	TGRS 2023	0.9711	36.9822	0.9796	0.0425
	PhDNet [28]	INFUSION 2024	0.9710	35.7371	0.9743	0.0466
	FmambaIR [46]	TGRS 2025	0.9632	33.9669	0.9691	0.0476
	KA-Net [56]	TPAMI 2025	0.9489	32.2984	0.9672	0.1331
	CloudRuler [45]	RSE 2025	0.9561	34.7169	0.9713	0.0380
	MP_HSIR* [48]	ICCV 2025	0.9664	37.0868	0.9723	0.0387
Proposed	**HTP-SSDN**	-	**0.9730**	**37.1567**	0.9808	**0.0290**

* denotes hyperspectral image dehazing or reconstruction methods

TABLE II

QUANTITATIVE COMPARISON OF DEHAZING METHODS ON THE HDD DATASET. BOLD AND UNDERLINED INDICATE BEST AND SECOND-BEST RESULTS

Methods		Venue	NIQE↓	AG↑
Prior-based	DCP [5]	TPAMI 2009	23.2737	0.1212
	Defog* [38]	TGRS 2022	20.7466	0.0671
Learning-based	FFANet [19]	AAAI 2020	17.8467	0.1983
	RSDehazeNet [20]	TGRS 2021	18.6571	0.1357
	Restormer [44]	CVPR 2022	15.9563	0.2102
	SGNet* [31]	ISPRS 2022	16.3662	0.2032
	PSMBNet [55]	TGRS 2023	17.9716	0.1470
	DehazeFormer [23]	TIP 2023	16.3873	0.2456
	AIDTransformer [24]	CVPR 2023	15.6097	0.2330
	AACNet* [33]	TGRS 2023	15.2740	**0.2556**
	PhDNet [28]	INFUSION 2024	16.8320	0.2371
	FmambaIR [46]	TGRS 2025	16.7542	0.2395
	KA-Net [56]	TPAMI 2025	17.3059	0.2246
	CloudRuler [45]	RSE 2025	15.4150	0.2503
	MP_HSIR* [48]	ICCV 2025	15.0658	0.2522
Proposed	**HTP-SSDN**	-	**14.9563**	0.2544

* denotes hyperspectral image dehazing or reconstruction methods

traditional CNN method. Defog outperformed DCP in dehazing metrics, demonstrating its capability in dense haze elimination; the HSI dehazing method generally outperformed both natural and multispectral dehazing methods. For instance, AACNet achieved the highest AG, while MP_HSIR attained the suboptimal NIQE, further validating the necessity of designing specialized HSI dehazing approaches. The proposed HTP-SSDN achieved the optimal NIQE and suboptimal AG, demonstrating its effectiveness in HSI dense haze removal.

To further demonstrate the scene recovery ability of different dehazing methods, Fig. 7 shows six scenes with dense haze, with the red and blue zooming showing the detailed comparison. According to the results, it can be seen that defog had a better dehazing effect on some scenes (regions 2 and 3) than learning-based methods, but there were obvious spectral aberrations in the reconstruction of the scenes (regions 1, 4, and 5), and thus its quantitative index was poorer. RSDehazeNet, DehazeFormer, AIDTransformer, SGNet, and MP_HSIR all suffered from incomplete removal of dense haze (regions 5 and 6). AACNet removed most of the haze in regions 2, 3, 4, and 5, but the recovered surface scene had some distortions, and its haze removal was incomplete in regions 1 and 6; in contrast, the proposed methods all effectively recovered clean scene information, and the visual effect was better than AACNet with proximate metrics.

Real-World Challenging Scenes: To evaluate the generalization performance of different hyperspectral dehazing methods in real-world scenes, a snowy landscape characterized by varying haze densities was selected for validation, with the visual results depicted in Fig. 8. Given the high visual similarity between snow and haze, this scene effectively tests a method's capability for haze mitigation. As illustrated in Fig. 8, the model-based method, defog, only manages to eliminate thin haze. SGNet suffers from residual haze and noticeable color distortion under moderate and thick haze. While MP_HSIR and AACNet successfully suppress thin-to-moderate haze, they both fail in thick haze regions. On the contrary, the proposed method effectively restores the scenes across all haze levels while faithfully preserving the white snow background, validating its robustness and effectiveness in highly challenging real-world environments.

C. Spectral Reconstruction Results

In this section, the underhaze spectra of regions 5 and 6, as shown in Fig. 6, were selected to evaluate the spectral reconstruction performance of various dehazing methods, as shown in Fig. 9. It can be seen that the haze spectral curves

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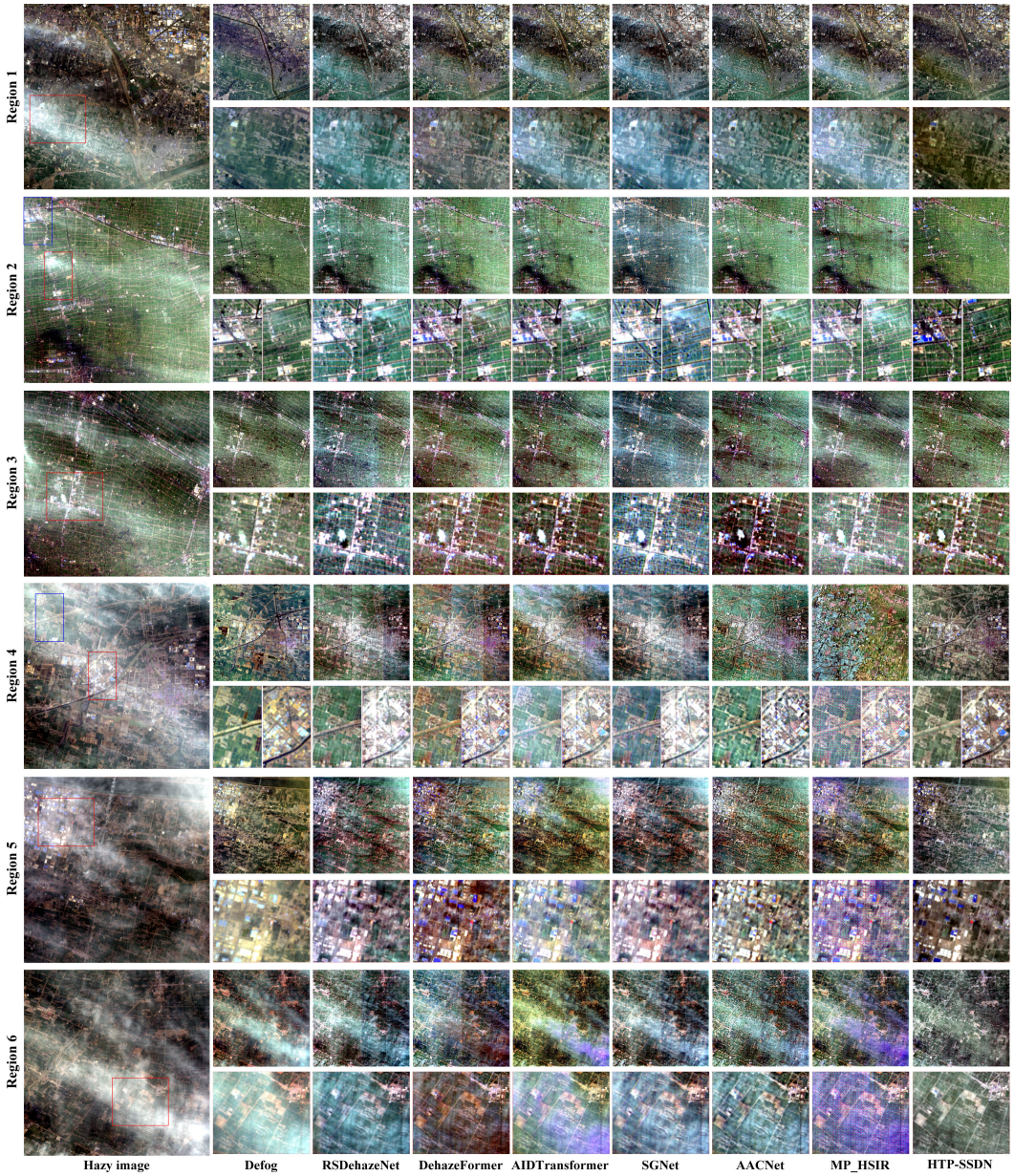


Fig. 7. Dehazing results (Red: band 59, Green: band 38, Blue: band 20) of various methods on the HDD dataset. For each region, (top row) dehazing result for the whole scene, and (bottom row) zoomed area.

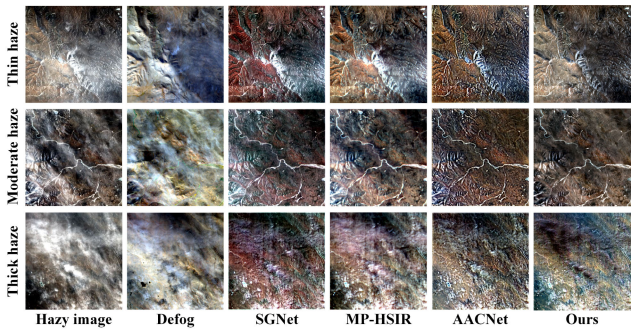


Fig. 8. Dehazing results (Red: band 59, green: band 38, and blue: band 20) of various HSI dehazing methods on a snowy landscape with varying haze levels.

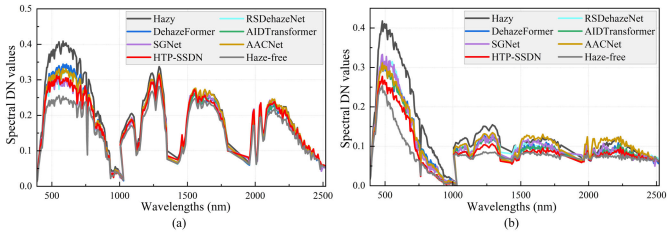


Fig. 9. Spectrum reconstruction performance on the HyperDehazing from hazy HSIs. (a) Region 5-(320, 170). (b) Region 6-(400, 423).

were significantly higher than the reference curves without haze, and most of the methods can reduce the DN values in the visible spectrum, but the performance varies. In particular, RSDehazeNet, DehazeFormer, and AIDTransformer had poorer haze removal performance in the visible spectrum; SGNet had better spectral recovery in region 5 but poorer

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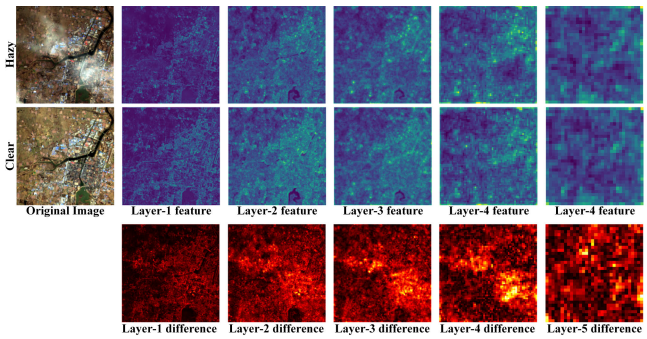


Fig. 10. Multilevel features and their differences extracted from hazy images and clear images by the pretrained VGG19 network, with significant feature differences demonstrating the effectiveness of MCR.

TABLE III

COMPARISON OF HTP-SSDN USING DIFFERENT PRETRAINED MODELS AS CONTRASTIVE REGULARIZATION ON THE HYPERDEHAZING DATASET. TRAINING TIME INDICATES THE AVERAGE TRAINING TIME PER ITERATION. (500 ITERATIONS)

Pre-trained model	SSIM↑	PSNR↑	UQI↑	SAM↓	Training time(s)
ResNet50	**0.9705**	35.2716	**0.9707**	0.0319	8.60
ViT-B_16	0.9663	**35.8649**	0.9667	0.0380	7.55
SA+VGG19	0.9667	34.5842	0.9600	0.0329	7.21
VGG19	0.9678	35.5955	0.9644	**0.0313**	**7.01**

results in region 6; and AACNet performed well in the visible wavelength range but produced higher spectral profiles in the infrared spectrum. Overall, the spectral profiles achieved by the proposed HTP-SSDN were closer to the reference (region 6), and the spectral trends were highly consistent despite some differences (region 5).

D. Contrast Regularization Analysis

To validate the rationality and effectiveness of MCR, this section compared VGG19 with other pretrained encoders: spectral adapter (SA) [57] plus VGG19, which accepts full-spectrum HSI without band selection; ResNet50 (four feature layers); and ViT-B_16 (four token layers reshaped into feature maps). Results are reported in Table III. Compared with VGG19, SA's use of a full-spectrum input introduced long-wavelength channels where hazy and clear images differ little, degrading both dehazing performance and training efficiency; ResNet50 yielded modest performance gains at the cost of a longer training time; ViT-B_16 achieved the best PSNR. This demonstrated that multilevel visual features accelerate contrastive regularization convergence, and appropriate models can be selected based on requirements. Although VGG19 was suboptimal on the first three metrics, its advantages in spectral reconstruction and computational efficiency motivated its use for HSI dehazing. Fig. 10 shows multilayer feature comparisons between hazy and clean images using VGG19: distinct differences appear across layers. High-level semantic features help assess output correctness and, via guided gradient descent, drive the network output closer to the clean reference and away from the hazy input.

TABLE IV

ABLATION ANALYSIS OF DIFFERENT MODULES IN HTP-SSDN ON THE HYPERDEHAZING DATASET. (500 ITERATIONS)

Base	SpaRM	SpeRM	PFAM	HTP	SSIM↑	PSNR↑	UQI↑	SAM↓
✓					0.8880	30.9869	0.9493	0.0803
✓	✓				0.9452	32.5014	0.9571	0.0513
✓		✓			0.9230	31.9523	0.9517	0.0487
✓	✓	✓			0.9556	33.8963	0.9588	0.0394
✓			✓		0.8891	31.1962	0.9501	0.0784
✓			✓	✓	0.9326	33.1897	0.9546	0.0525
✓	✓	✓	✓	✓	**0.9678**	**35.5955**	**0.9644**	**0.0313**

TABLE V

ABLATION ANALYSIS OF DIFFERENT LOSSES IN HTP-SSDN ON THE HYPERDEHAZING DATASET. (500 ITERATIONS)

$\mathcal{L}_1$	$\mathcal{L}_{spe}$	$\mathcal{L}_{MCR}$	SSIM↑	PSNR↑	UQI↑	SAM↓
✓			0.9583	32.6739	0.9601	0.0659
✓	✓		0.9607	34.2133	0.9617	0.0343
✓		✓	0.9633	35.0851	0.9605	0.0418
✓	✓	✓	**0.9678**	**35.5955**	**0.9644**	**0.0313**

TABLE VI

UNIQUENESS ANALYSIS OF HTP INJECTION ON THE HYPERDEHAZING DATASET. (500 ITERATIONS)

Prior-injection	SSIM↑	PSNR↑	UQI↑	SAM↓
DPC	0.9342	33.2970	0.9460	0.0421
SAF	0.9623	35.0791	0.9586	0.0362
SHI	0.9511	34.4026	0.9523	0.0397
HTP	**0.9678**	**35.5955**	**0.9644**	**0.0313**

E. Ablation Studies

In this section, the network modules and loss functions of HTP-SSDN were ablated on the HyperDehazing dataset to validate the performance of different modules through 500 learning iterations, as presented in Tables IV–VI.

Ablation Analysis of Different Modules: The base was an end-to-end dehazing network containing only convolutional layers. Table IV presents that the base dehazing performance was limited. Both SpaRM and SpeRM improved all metrics by enhancing spatial and spectral representations, respectively: SpaRM yielded larger gains on the first three metrics, while SpeRM contributed more to SAM. Their combination achieved suboptimal overall results, demonstrating complementarity and effectiveness. The standalone PFAM added only a few convolutional layers, so its individual improvement was modest; however, when combined with HTP, it injected useful priors and substantially boosted network performance. The effective combination of these modules consistently improved dehazing capability, facilitating accurate detection of hazy regions and faithful reconstruction of scene details under haze.

Ablation Analysis of Different Losses: The different dehazing loss functions were further evaluated. Table V presents the combined metrics of $\mathcal{L}_1$, $\mathcal{L}_{spe}$, and $\mathcal{L}_{MCR}$. The $\mathcal{L}_1$ loss guided the network to better performance in most of the metrics by constraining the dehazed and reference images; $\mathcal{L}_{spe}$ further constrained the reconstruction of the spectrum and obtained a better SAM, and the addition of $\mathcal{L}_{MCR}$ increased the contrast

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TABLE VII
ANALYSIS OF THE EFFECT OF L IN SSDB. (500 ITERATIONS)

L	SSIM↑	PSNR↑	FLOPs (G)	Params (M)
2	0.9668	35.4116	**3.32**	**2.79**
3	0.9678	**35.5995**	3.85	3.56
4	0.9684	35.4509	4.37	4.74
5	**0.9685**	35.5898	4.90	5.71

TABLE VIII
ANALYSIS ON THE EFFECT OF α IN SPERM. (500 ITERATIONS)

α	SSIM↑	PSNR↑	FLOPs (G)	Params (M)
1/8	0.9641	35.6440	**3.59**	3.70
1/4	0.9634	35.3773	3.67	3.72
1/2	**0.9678**	35.5995	3.85	**3.56**
3/4	0.9669	36.1927	4.05	3.81
7/8	0.9672	**36.5001**	4.16	3.84

with the hazy image and further helped in generating a better recovered image. Overall, the combination of the three losses was effective in HSI dehazing.

Uniqueness Analysis of HTP: To validate HTP, we compare it against three alternatives: direct prior concatenation (DPC) (where the prior and image are used as joint inputs); spatial attention fusion (SAF); and single-stage HTP injection (SHI), as presented in Table VI. DPC yields marginal gains as it treats the prior merely as an auxiliary input. SHI fails to represent haze features in deeper layers due to its shallow-only injection. While SAF integrates priors via attention, it often overlooks intrinsic image structures, leading to suboptimal restoration. On the contrary, our HTP consistently outperforms these variants, demonstrating superior adaptability in fusing haze priors across multiscale features.

F. Parameter Sensitivity Analysis

This section evaluated the impact of several key hyperparameters on dehazing performance, including: the number of SSRM-PFAM-SSRM in SSDB L, separation coefficient α, λ_1, and λ_2 in the loss function, and batch size.

The parameter L determined the performance of spectral-spatial refinement feature extraction and scene reconstruction in HTP-SSDN. The sensitivity analysis of L is presented in Table VII. It can be seen that the larger L improved SSIM, FLOPs, and the number of parameters, but PSNR fluctuated, which indicated that a deeper SSDB module may bring in features that are not helpful for dehazing. To strike a balance between performance and computational cost, L was set to 3.

To evaluate α in SpeRM, we varied it from 1/8 to 7/8, as presented in Table VIII. Increasing α led to progressive gains in PSNR and FLOPs, indicating that larger splits retained richer features during the SpeRM transformation and improved overall performance. $\alpha = 1/2$ yielded the highest SSIM with fewer parameters, making it the optimal choice for balancing performance and efficiency.

Fig. 11 showed the effect of different combinations of λ_1 and λ_2 on dehazing performance. Because the constituent losses have different numerical scales, their weights should

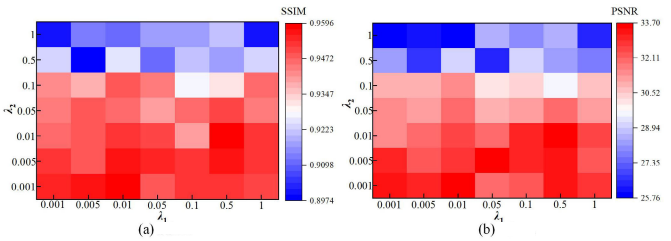


Fig. 11. Analysis of different combinations of λ_1 and λ_2 in the loss function on (a) SSIM and (b) PSNR.

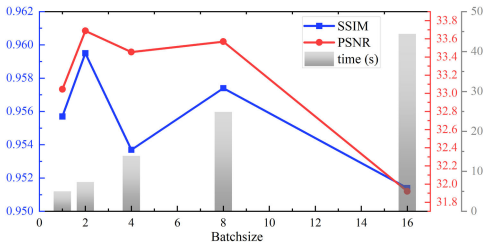


Fig. 12. Analysis of batch size on SSIM, PSNR, and the average training time.

be normalized accordingly. The results indicated that λ_2 had a larger impact on performance, and a smaller λ_2 improved dehazing capability. We therefore set $\lambda_1 = 0.01$ and $\lambda_2 = 0.001$ to obtain the best results.

Fig. 12 shows the comparison of dehazing results for different batch sizes. Larger batch sizes did not necessarily improve performance but substantially increased training time. We therefore used a batch size of 2 to achieve the best tradeoff between performance and efficiency.

G. Model Parameters and Efficiency

In this section, we evaluated the number of hyperparameters and FLOPs for HTP-SSDN and other dehazing networks under the same experimental settings, which are presented in Table IX. It can be seen that most transformer-based dehazing networks exhibited high parameters and FLOPs, whereas CloudRuler leveraged physical constraints to substantially reduce computational complexity and improve efficiency. For HSI dehazing, AACNet and MP HSIR were relatively complex, while SGNet achieved lower complexity at the cost of markedly reduced performance. The proposed HTP-SSDN contained only 3.56 M parameters, with FLOPs and training time among the second best, and inference time also near the top in HSI dehazing methods. Excluding the HTP module further reduces parameters and FLOPs. While the MCR loss imposes no additional parameter overhead, it marginally increases computational costs during training. Ultimately, the proposed framework achieves an optimal tradeoff between dehazing accuracy and architectural efficiency.

H. Downstream Task Application

To assess the impact of HSI dehazing on downstream tasks, we evaluated its performance across two distinct domains: terrestrial land cover classification and marine oil spill detection.

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TABLE IX

COMPARATIVE ANALYSIS OF DEHAZING METHODS: PARAMS, FLOPS, AVERAGE TRAINING TIME PER ITERATION, AND AVERAGE INFERENCE TIME PER IMAGE. BOLD AND UNDERLINED INDICATE BEST AND SECOND-BEST RESULTS. (PERFORMANCE RANKINGS ARE INDICATED IN PARENTHESES)

Methods	Params (M)	FLOPs (G)	Training time (s)	Inference time (s)
CANet [41]	12.12	12.96	7.06	5.41
Restormer [44]	101.16	10.43	9.33	19.11
LKDMNet [42]	11.06	**2.24**	8.48	7.83
DehazeFormer [23]	10.24	7.48	8.09	3.91
AIDTransformer [24]	125.11	30.76	9.43	78.77
PhDNet [28]	4.78	5.44	8.42	4.46
FmambaIR [46]	4.20	15.33	8.20	10.06
KA-Net [56]	56.63	11.03	8.31	**1.93**
CloudRuler [45]	**1.91**	6.00	**6.78**	7.67
SGNet* [31]	4.32	4.52	7.21	<u>3.16</u>
AACNet* [33]	12.76	13.18	7.36	10.08
MP_HSIR* [48]	152.9	51.36	8.96	26.02
HTP-SSDN	<u>3.56</u> (2)	<u>3.85</u> (2)	<u>7.01</u> (2)	6.84 (6)
HTP-SSDN w/o HTP	2.79	3.16	6.57	5.32
HTP-SSDN w/o MCR	3.56	3.85	6.49	5.01

* denotes hyperspectral image dehazing or reconstruction methods

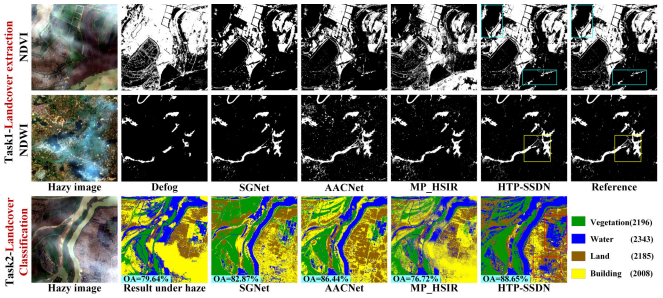


Fig. 13. Assessment of HSI dehazing effects on land cover applications. Task 1: vegetation and water extraction maps. Task 2: land cover classification results.

Terrestrial land cover classification includes: 1) recovery of typical surface features using the normalized difference vegetation index (NDVI) and the normalized difference water index (NDWI), and 2) land cover classification accuracy. Results are presented in Fig. 13. For task 1), identical thresholds were applied to both indices across all dehazing methods. Defog produced extensive false vegetation and large water omission (yellow rectangle). SGNet missed substantial vegetation and water regions (yellow and blue rectangles). AACNet tended to overextract water, introducing noise. MP_HSIR misclassified water as vegetation (blue rectangles). The proposed HTP-SSDN most closely matched the reference. For task 2), we manually selected samples for four classes—vegetation, water, land, and building—using ten random samples per class for training and the remaining samples for testing. A support vector machine (SVM) classifier [58] was used, and overall accuracy (OA) was reported. The hazy images exhibited severe misclassification (e.g., haze-covered land identified as building). SGNet and AACNet improved OA by 3.23% and 6.80%, respectively, although SGNet produced a vegetation misclassification. MP_HSIR failed to remove haze adequately, yielding a 2.92% decrease in OA. On the contrary, the HTP-SSDN achieved the highest classification accuracy and produced a class map most consistent with the true scene (red rectangles), confirming its superior dehazing capability and practical value.

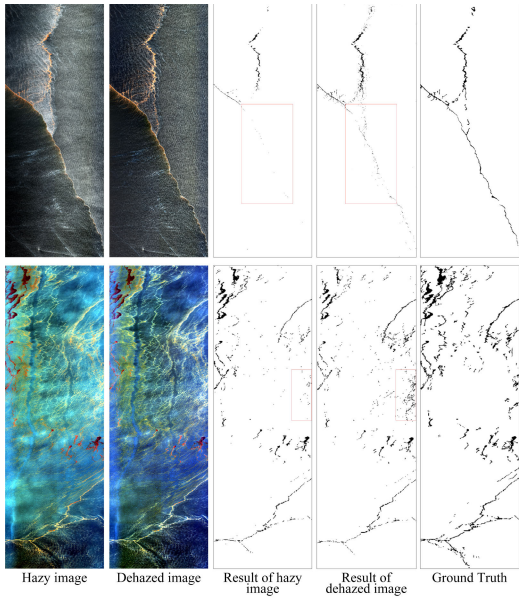


Fig. 14. Visualizations of detected oil leakage in various GM regions before and after dehazing using our HTP-SSDN.

For marine oil spill applications, two HSIs affected by sea haze (GM02 and GM05) were selected from the Gulf of Mexico (GM) oil spill dataset [59]. We evaluated both the dehazing performance of the proposed HTP-SSDN and its subsequent impact on oil spill detection using the rank-two nonnegative matrix factorization (R2NMF) [60] detector, as illustrated in Fig. 14. As observed, the presence of sea haze severely degrades detection performance, causing R2NMF to suffer from significant missed detections of oil spill regions (indicated by the red rectangles). Following dehazing with our HTP-SSDN, the R2NMF detector achieves a substantially higher detection accuracy, yielding results that align much more closely with the ground truth. This finding strongly validates the practical deployment potential of the proposed method.

I. Generalization Ability on MSIs Data

In this section, the generalization dehazing performance of the proposed HTP-SSDN was evaluated on the Landsat-8 and Sentinel-2 MSI datasets, which were widely used for haze and cloud removal. In this case, bands 1–7 of Landsat-8 and bands 2, 3, 4, and 8 of Sentinel-2 were used, and the corresponding visible and infrared bands were used to construct the HTP; their quantitative analyses are presented in Table X. It can be seen that the two transformer-based models, DehazeFormer and AIDTransformer, achieved the best results in some metrics. HTP-SSDN achieved the best SSIM, PSNR, and SAM on Landsat-8, and the best SSIM, SAM, and suboptimal PSNR on Sentinel-2. This further demonstrates the

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TABLE X
COMPARISON OF QUANTITATIVE RESULTS ON LANDSAT-8 AND SENTINEL-2. BOLD AND UNDERLINED INDICATE BEST AND SECOND-BEST RESULTS

Dataset	Landsat-8				Sentinel-2			
Methods	SSIM↑	PSNR↑	UQI↑	SAM↓	SSIM↑	PSNR↑	UQI↑	SAM↓
PSMBNet [55]	0.9350	25.1137	0.8357	0.0318	0.9073	30.1267	0.9073	0.0837
DehazeFormer [23]	0.9176	<u>25.8434</u>	**0.8420**	0.0198	0.9107	30.0420	**0.9289**	0.0579
AIDTransformer [24]	0.9468	25.6179	0.8398	0.0208	0.9167	**30.5063**	<u>0.9250</u>	0.0749
SGNet [31]	0.9412	25.3269	0.8233	0.0203	0.8894	28.2104	0.8843	0.0961
AACNet [33]	0.9461	25.4067	0.8239	0.0193	<u>0.9157</u>	30.1584	0.9215	0.0746
CloudRuler [45]	<u>0.9485</u>	25.7951	<u>0.8419</u>	<u>0.0179</u>	<u>0.9169</u>	30.2516	0.9234	<u>0.0562</u>
HTP-SSDN	**0.9492**	**25.8646**	0.8415	**0.0169**	**0.9170**	<u>30.3016</u>	0.9224	**0.0534**

excellent general ability of the proposed method for remote sensing image dehazing and scene reconstruction.

V. CONCLUSION

This article presents an HSI dehazing method, HTP-SSDN, combining haze prior and deep networks. The proposed HTP and PFAM inject the haze prior into the multilayer features of the network, which improves the network's ability to sense haze and reconstruct the scene under haze. The presented SSDM significantly improves the performance of scene detail and spectral reconstruction by extracting refined spectral and spatial features. In addition, the introduced contrast regularization with positive sample constraints further improves the network's learning and dehazing capabilities. Extensive experimental results demonstrate the effectiveness, efficiency, and generality of HTP-SSDN in remote sensing image dehazing. In future work, we will focus on the development of methods for weakly supervised, zero-shot, and multimodal remote sensing data to enhance scene dehazing under missing image pairs.

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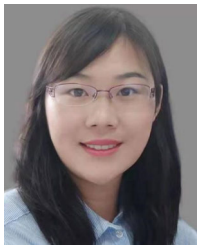
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